Measuring Fix-It: Gaps and Overlaps
Answer Key
Grade 1

Quick Answers

1. 3 cubes
2. (a) clips the gaps hid = 3, —
3. 8 sticks, —
4. 7 paper clips, —
5. <
6. (a) Kim's fixed count = 12, —

What is verified

2 of 6 problems fully machine-checked against SymPy.
— marks an answer only you can judge — problems 2, 3, 4, 6. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 1
1.MD – *problems 1, 2, 3, 4, 5, 6*
Difficulty 1–4 of 5 across 10 tagged checks.

Common wrong answers

2. *If they got 6: used the gap count as if it were the true length*
3. *If they got 10: kept the overlapped count instead of taking the extras away*
4. *If they got 4: kept the count with gaps instead of adding the hidden clips*
6. *If they got 15: kept the overlapped count*

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1. Pencil: 8 cubes. Crayon: 5 cubes. How many cubes longer is the pencil?

Solution: Both lengths use the same unit (cubes), so subtract the counts: $8 - 5 = 3$. The pencil is longer by $\boxed{\text{3 cubes}}$

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2. Mia's ribbon: 6 clips with gaps, 9 clips with no gaps. (a) How many clips did the gaps hide? (b) Which count is true?

Solution: (a) The difference between the two counts is what the gaps hid: $9 - 6 = 3$. $\boxed{3}$

(b) *Instructor judges:* the 9-clip count is the true length. Gaps skip over parts of the ribbon, so a gappy count is always too small.

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3. Leo overlapped sticks: his count of 10 is 2 too many. Fix it and tell what went wrong.

Solution: Overlapping counts the same part of the table twice, so take the extra sticks away: $10 - 2 = 8$. The table is really $\boxed{\text{8 sticks}}$

Tell what went wrong (instructor judges): accept any wording like “Leo piled sticks on top of each other, so he counted some length twice. His number is too big, and we take the 2 extras away.”

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4. Zoe left 3 gaps, each gap exactly 1 clip of rope, and counted 4 clips. Fix it and tell how you know.

Solution: Each gap is rope that never got a clip, so add the hidden clips back: $4+3 = 7$. The rope is really $\boxed{\text{7 paper clips}}$

Tell how you know (instructor judges): accept any wording like “the 3 gaps each hid 1 clip of rope, so Zoe’s 4 is missing 3 clips. Gaps make a count too small.”

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5. Blue string 11 clips, red string 14 clips. Write >, <, or =: 11 __ 14.

Solution: 11 clips is fewer than 14 clips, so the blue string is shorter: $\boxed{11 < 14}$

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6. Kim overlapped blocks: her count of 15 is 3 too many. Ali counted 12 with no overlaps. (a) Fix Kim’s count. (b) Do they agree now?

Solution: (a) Take away the 3 blocks Kim counted twice: $15 - 3 = 12$. $\boxed{12}$

(b) *Instructor judges:* yes — Kim’s fixed count is 12 blocks, exactly Ali’s count. Two careful measurements of the same bench with the same unit should match.